AMR MANSOUR MUHAMMAD RASHAD

Road Construction Engineer | Data Analyst | PhD Candidate – Highway & Airport Engineering

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PROFESSIONAL SUMMARY

- Road Construction Engineer at the Egyptian Ministry of Transport (General Authority for Roads and Bridges), supervising national road construction and maintenance projects (50+ km, EGP 500M+ BOQs).
- PhD Candidate in Highway and Airport Engineering (Aswan University) with 2 peer-reviewed publications on slope stability and soil nailing.
- Data Analytics skills: SQL, Python (Pandas), Excel, Power BI, Tableau, Statistics, applied to pavement and infrastructure datasets.
- Former Assistant Lecturer and Teaching Assistant at Nahda University (5+ years) across 10+ civil engineering courses.

KEY SKILLS

Technical: Highway & Road Engineering, Pavement Design & Materials, Slope Stability, Soil Nailing, BOQ & Quantity Estimation, Contract Documentation, Tender Evaluation, Quality Control, BIM, Geotechnical & Numerical Analysis

Data & Digital: Data Analytics, Data Visualization, Advanced Excel, SQL, Python (Pandas), Power BI, Tableau, SPSS, Statistics, Data Mining, Predictive Analytics, Machine Learning, KPI Dashboards

Software: AutoCAD, Civil 3D, InfraWorks, Plaxis 2D, GeoStudio, MATLAB, ETABS, SAP2000, Primavera, Microsoft Office

Soft Skills: Project Management, Planning, Leadership, Teamwork, Crisis Management, Presentation

PROFESSIONAL EXPERIENCE

Road Construction Engineer *Sep 2024 – Present*

Ministry of Transport – General Authority for Roads and Bridges (Central Administration for Roads Construction and Maintenance) | Cairo, Egypt

- Supervise and manage the execution of national road construction and maintenance projects covering over 50 km of highways, achieving 100% compliance with MoT specifications.
- Approve BOQs and quantity take-offs for projects with a combined value exceeding EGP 500,000,000.
- Evaluated 10+ tenders, contributing to a 15% faster contractor selection and contract award process.
- Reduced on-site technical non-conformances by 20% through systematic on-site QC inspections.
- Prepare technical specifications, quantity estimates, contract documentation and progress reports, improving documentation efficiency by 30%.

Data Analytics Trainee *Sep 2026 – Present*

National Telecommunication Institute (NTI) | Giza, Egypt

- Completing an intensive data analytics program covering SQL, Python (Pandas), Excel, Power BI, Tableau and Statistics, building portfolio-ready projects from data collection through visualization and business reporting.

Data Analytics Trainee – Digital Egypt Pioneers Initiative (DEPI) *Nov 2025 – Jul 2026*

Ministry of Communications and Information Technology (MCIT) | Cairo, Egypt

- Developed 3 interactive Power BI dashboards that increased KPI visibility for transportation and infrastructure datasets by 40%.
- Analyzed datasets of 1,000+ records on pavement performance, improving data accessibility for research by 50%.
- Trained in data mining, visualization and statistical analysis to support infrastructure decision-making and highway research.

AI Ambassador Intern *Apr 2025 – May 2025*

National Telecommunication Institute (NTI) in collaboration with Engineers for a Sustainable Egypt (ESE) | Cairo, Egypt

- Selected for a program applying AI and digital transformation to support sustainable development goals in the engineering sector, covering 5 sustainable engineering development goals.

Engineering Trainee – Internship Program *Mar 2024 – Aug 2024*

Egyptian Military Academy | Cairo, Egypt

- Completed Training Course No. (1) for newly appointed engineers (General Authority for Roads, Bridges and Land Transport), ranking in the top 10% of participants.
- Strengthened skills in project management, technical reporting (accuracy improved by 25%), leadership, communication and time management.

Assistant Lecturer – Civil Engineering *Sep 2022 – Oct 2023*

Nahda University (NUB) | Bani Suwayf, Egypt

- Conducted 500 laboratory and practical sessions across 10 subjects (Fluid Mechanics, Hydraulics, Surveying, Soil Mechanics, Highway Materials), increasing student engagement by 35%.
- Trained 500 undergraduate students on Civil 3D, Excel and Primavera, achieving a 90% satisfaction rate.
- Assisted in teaching 10 civil engineering courses, including Highway Engineering, Transportation & Traffic Engineering, Public Works Projects, Project Management and Sanitary & Environmental Engineering, with a 100% attendance and material delivery record.
- Supervised 30 students during a 1-month summer training program at the National Research Center.

Head of Scheduling Committee – Civil Engineering Department *Mar 2019 – Sep 2022*
Nahda University (NUB) | Bani Suwayf, Egypt

- Led the development and coordination of academic schedules, aligning them with departmental and university calendars and resource availability.
- Optimized timetables for 50+ civil engineering courses, reducing scheduling conflicts by 40%.

Teaching Assistant – Civil Engineering *Oct 2018 – Sep 2022*
Nahda University (NUB) | Bani Suwayf, Egypt

- Supported instruction in Highway Engineering and other civil engineering subjects, assisted in laboratory and practical sessions, and provided academic guidance to about 400 students per year.

Honorary Member – Students Union, Faculty of Engineering & Computer Science *Nov 2016 – Nov 2017*
Nahda University (NUB) | Bani Suwayf, Egypt

- Contributed to student representation and supported academic, extracurricular and community outreach events.

Field Trainer (Internship) *Jun 2016 – Jul 2016*
Orascom Construction PLC | Cairo, Egypt

- Completed a 2-month internship in site supervision, documentation and quality control, ensuring 100% adherence to QC and safety protocols on large-scale projects.

EDUCATION

PhD, Highway and Airport Engineering – Aswan University *Dec 2022 – Present*

MSc, Highway and Airport Engineering – Minia University *Oct 2019 – Sep 2022*

- Thesis: Applicability of Optimized Numerical Models of Slope Stability of Road Embankments Adjacent to Water Channels under Traffic Load.

Pre-Master, Civil Engineering – Minia University *Sep 2018 – Sep 2019*

Bachelor of Engineering, Civil Engineering – Nahda University (NUB) *Sep 2013 – Sep 2018*

- Ranked 1st in class, GPA 3.95/4.00 (93.67%). Graduation Project: Design of Nahda Freeway – 98% (A+).

PUBLICATIONS

- Rashad, A.M. et al. "Effect of Soil Nailing on the Slope Stability of Road Embankments under Traffic Load Using the Optimized LEM." Journal of Advanced Engineering Trends, Vol. 42, No. 2, 2023.
- Rashad, A.M. et al. "Optimized Numerical Models of Slope Stability of Roads Embankment Using Limit Equilibrium Method." 2nd International Conference on Civil Engineering, pp. 318–329, 2021.

CERTIFICATIONS & AWARDS

- Digital Egypt Pioneers Program – Data Analyst Specialist (MCIT)
- Artificial Intelligence Ambassadors Program – National Telecommunication Institute (NTI)
- Training Course No. (1) for Newly Appointed Engineers – General Authority for Roads, Bridges / Ministry of Transport
- Conducting Laboratory Tests for Materials and Asphalt Laboratories
- Fundamentals of Digital Transformation – Minia University (Apr 2021)
- Statistical Analysis Using SPSS – Minia University (Nov 2020)
- Excellent Demonstrator Award – Faculty of Engineering, Nahda University (Sep 2019)
- Excellent Student Award (2nd Place, University-wide) – Nahda University (Jul 2017)

ACHIEVEMENTS & INTERNATIONAL EXPERIENCE

- Founded the "Be Best Engineering" blog, generating 190,000 page views.
- English Proficiency Exam (EPE) score 573, equivalent to TOEFL (Jul 2021).
- Exchange Student, Engineering Program – Al-Zaytoonah University of Jordan, Amman (Aug 2016).
- Participant, International Mobility Programme – Universiti Malaysia Perlis (UniMAP) (Aug–Sep 2015).

LANGUAGES

Arabic: Native / Bilingual | English: Professional Working Proficiency